

		$v_{\text{rms}} = 1.3 \times 10^5 \text{ m s}^{-1}$
13	A	Using $pV = nRT$ $(pV)_1 / T_1 = (pV)_2 / T_2$ $(1.0 \times 10^5 \times 0.050) / 300 = (5.0 \times 10^5 \times V_2) / 900$ $V_2 = 0.030 \text{ m}^3$
14	B	Velocity and displacement